WILP-AIML - Machine Learning

Assignment 2

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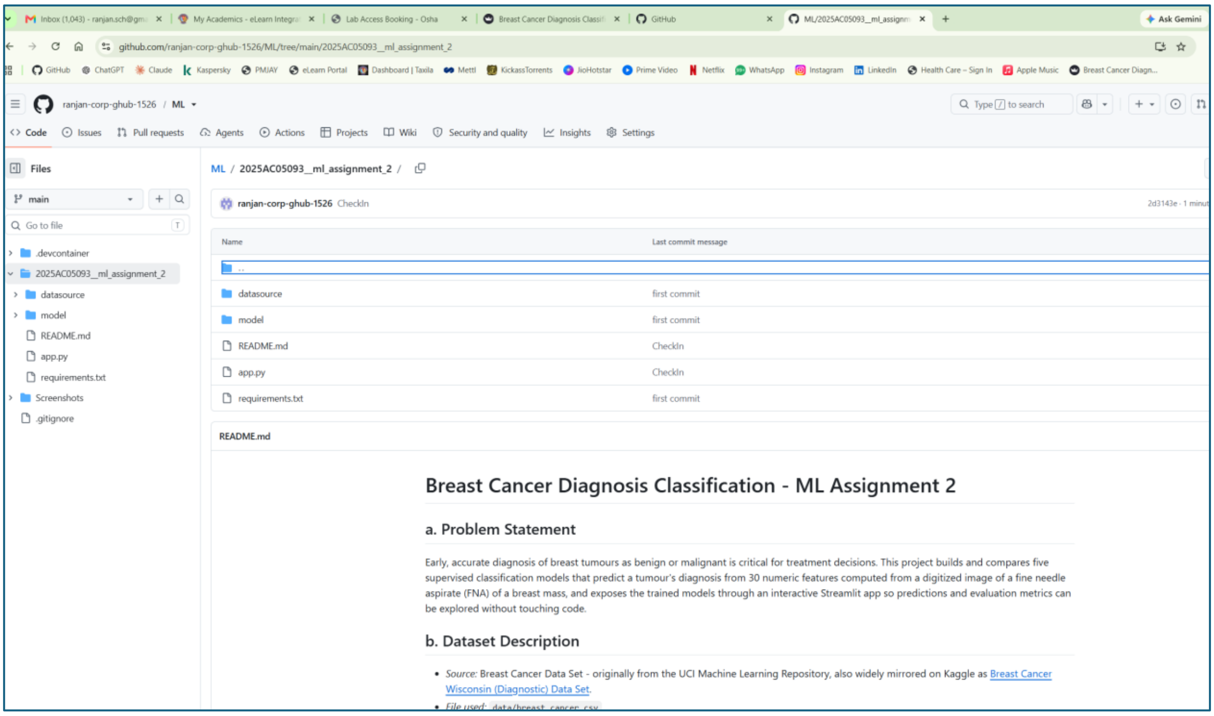
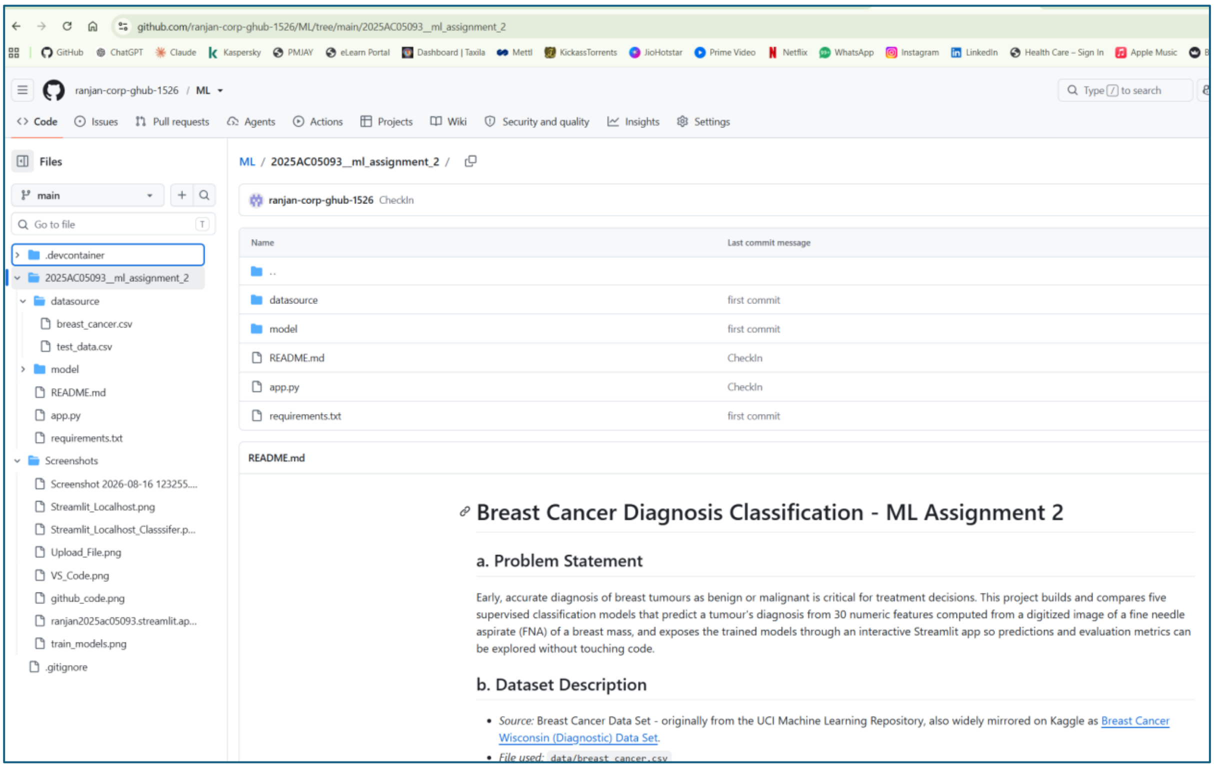
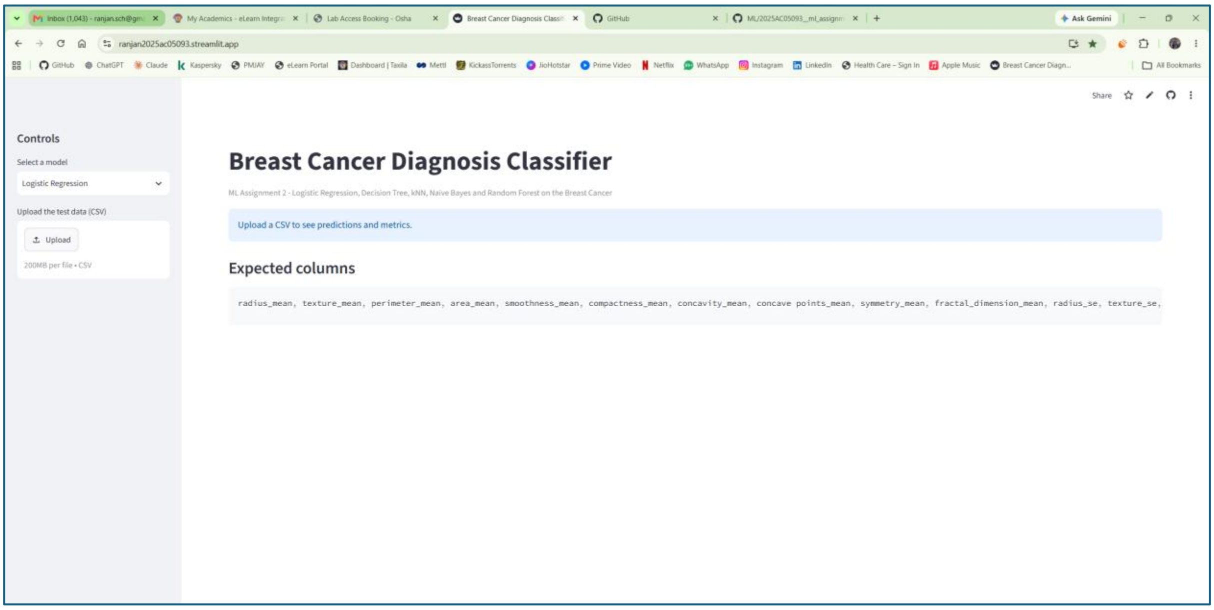
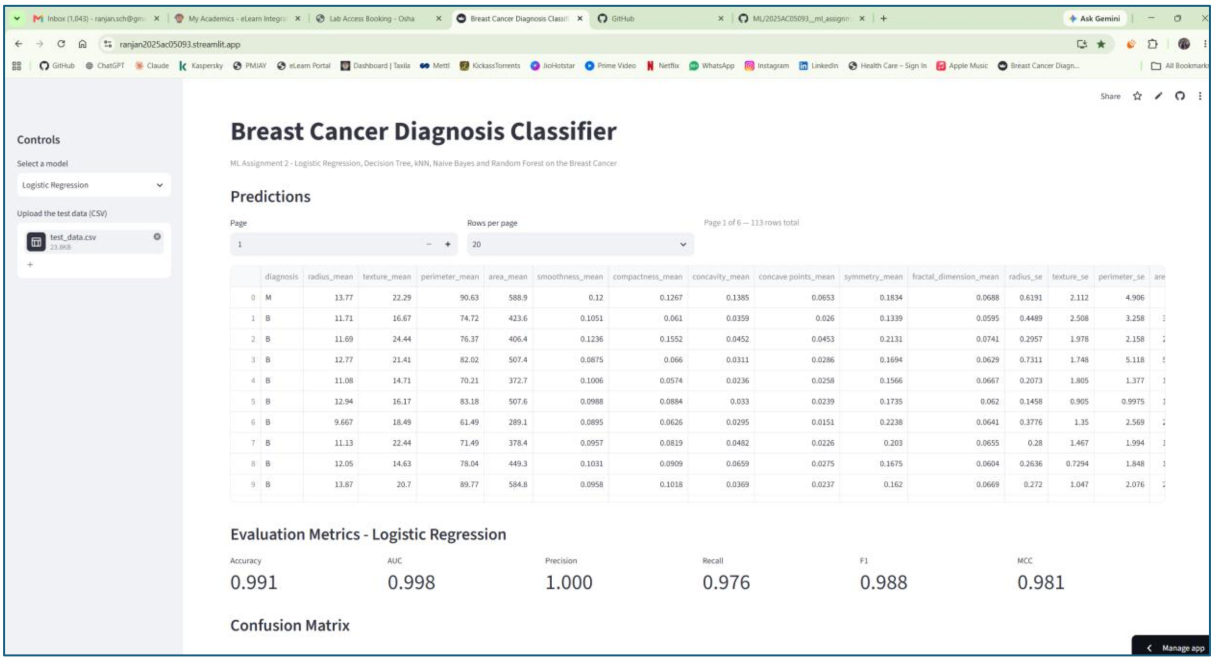
1. GitHub Repository Link – https://github.com/ranjan-corp-ghub-1526/ML



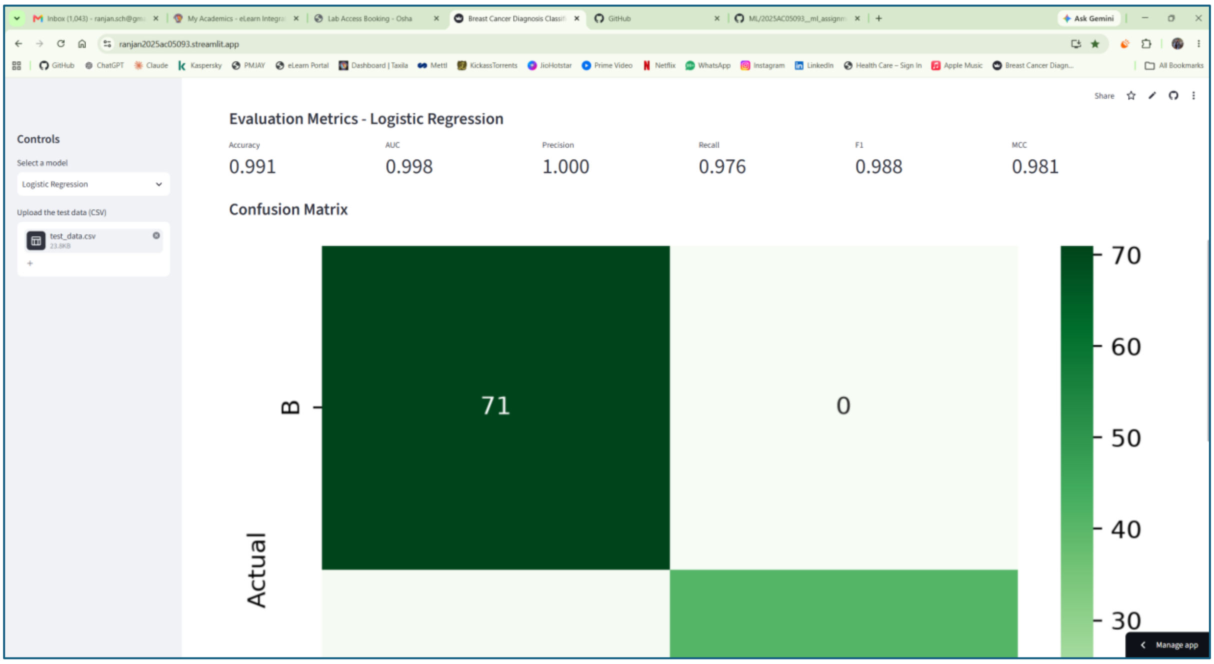


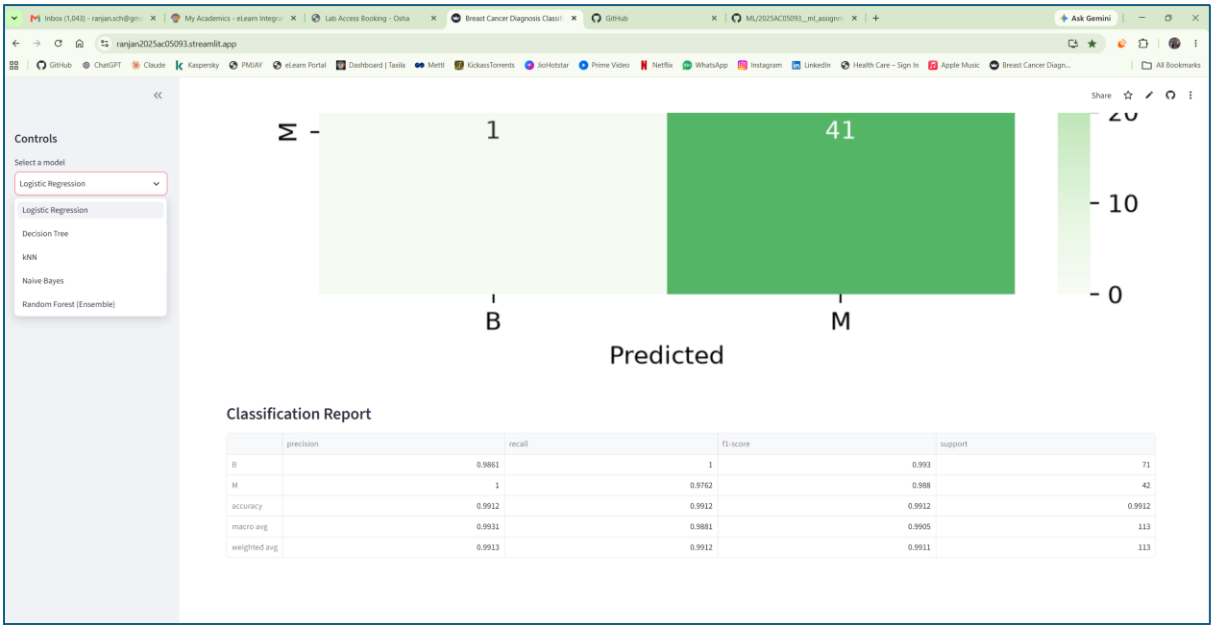
2. Live Streamlit App Link - https://ranjan2025ac05093.streamlit.app/



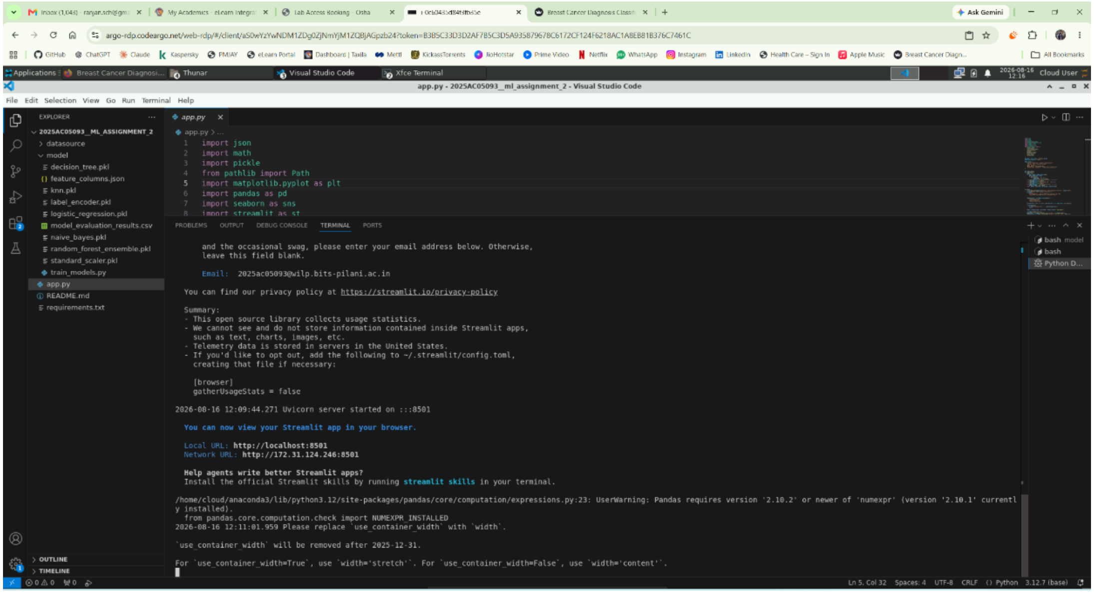


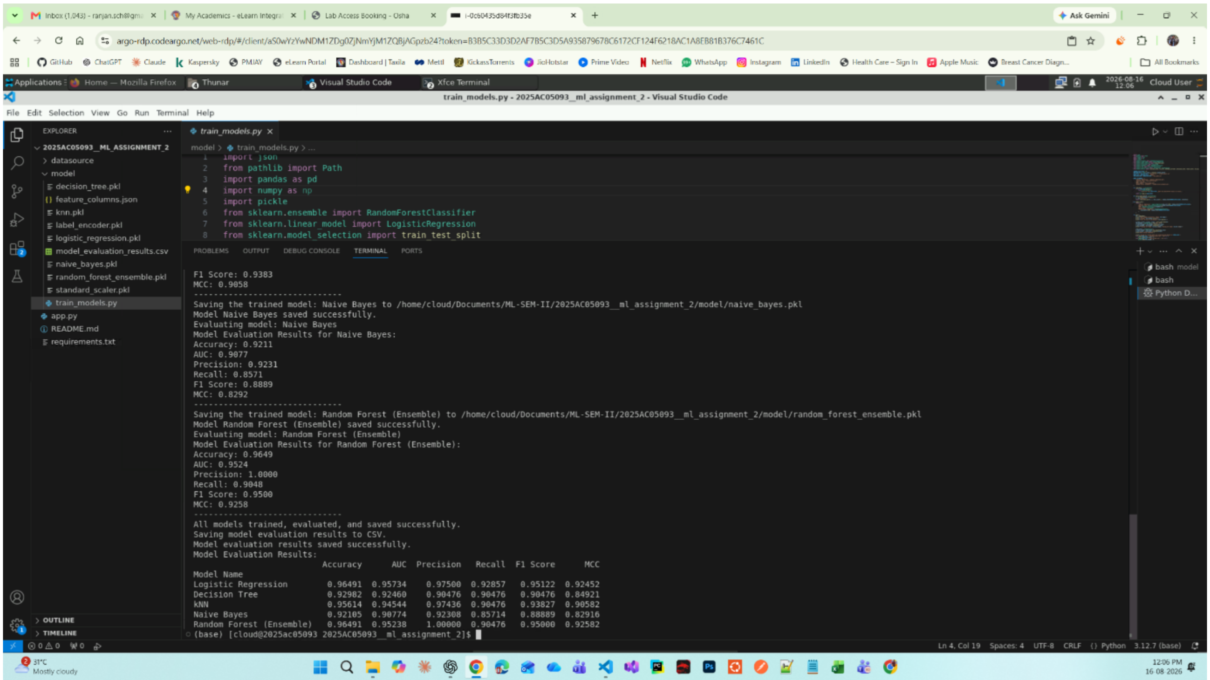
	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
0	M	13.77	22.29	90.63	588.9	0.12	0.1267	0.1385	0.0653	0.1834	0.0688	0.6191	2.112	4.906
1	B	11.71	16.67	74.72	423.6	0.1051	0.061	0.0359	0.026	0.1339	0.0595	0.4489	2.508	3.258
2	B	11.69	24.44	76.37	406.4	0.1236	0.1552	0.0452	0.0453	0.2131	0.0741	0.2957	1.978	2.158
3	B	12.77	21.41	82.02	507.4	0.0875	0.066	0.0311	0.0286	0.1694	0.0629	0.7311	1.748	5.118
4	B	11.08	14.71	70.21	372.7	0.1006	0.0574	0.0236	0.0258	0.1566	0.0667	0.2073	1.805	1.377
5	B	12.94	16.17	83.18	507.6	0.0988	0.0884	0.033	0.0239	0.1735	0.062	0.1458	0.905	0.9975
6	B	9.667	18.49	61.49	289.1	0.0895	0.0626	0.0295	0.0151	0.2238	0.0641	0.3776	1.35	2.569
7	B	11.13	22.44	71.49	378.4	0.0957	0.0819	0.0482	0.0226	0.203	0.0655	0.28	1.467	1.994
8	B	12.05	14.63	78.04	449.3	0.1031	0.0909	0.0659	0.0275	0.1675	0.0604	0.2636	0.7294	1.848
9	B	13.87	20.7	89.77	584.8	0.0958	0.1018	0.0369	0.0237	0.162	0.0669	0.272	1.047	2.076

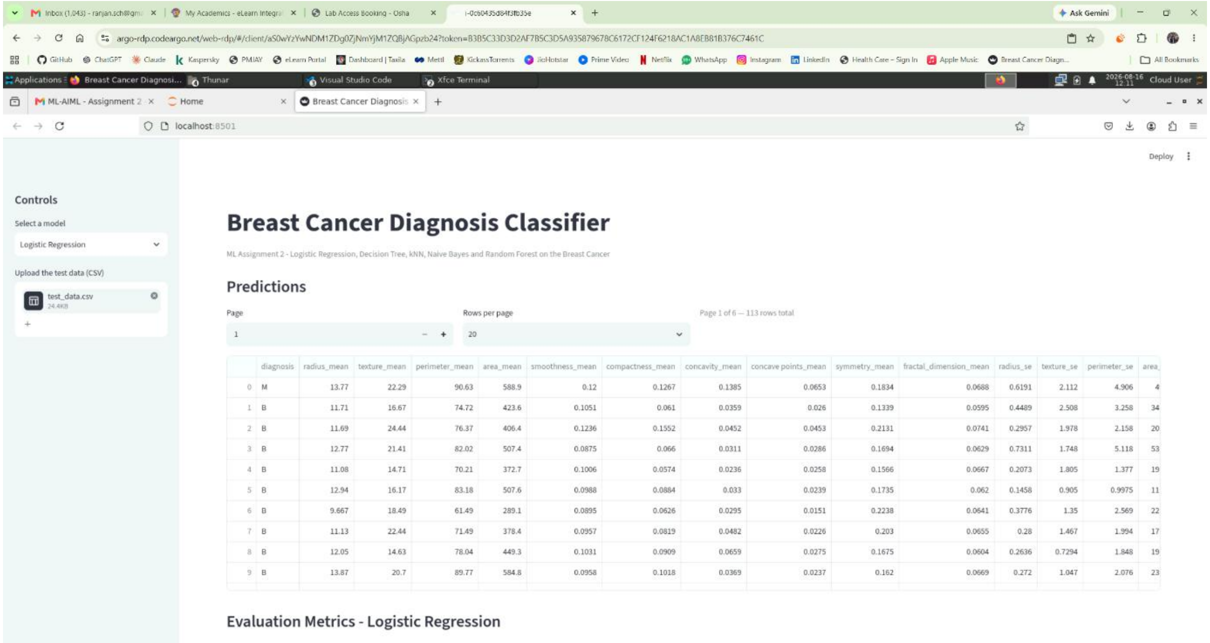




3. Screenshot







	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean	concavity_mean	concave points_mean	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se	area_
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2	B	11.69	24.44	76.37	406.4	0.1236	0.1552	0.0452	0.0453	0.2131	0.0741	0.2957	1.978	2.158	20
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4	B	11.08	14.71	70.21	372.7	0.1006	0.0574	0.0236	0.0258	0.1566	0.0667	0.2073	1.805	1.377	19
5	B	12.94	16.17	83.18	507.6	0.0988	0.0884	0.033	0.0239	0.1735	0.062	0.1458	0.905	0.9975	11
6	B	9.667	18.49	61.49	289.1	0.0895	0.0626	0.0295	0.0151	0.2238	0.0641	0.3776	1.35	2.569	22
7	B	11.13	22.44	71.49	378.4	0.0957	0.0819	0.0482	0.0226	0.203	0.0655	0.28	1.467	1.994	17
8	B	12.05	14.63	78.04	449.3	0.1031	0.0909	0.0659	0.0275	0.1675	0.0604	0.2636	0.7294	1.848	19
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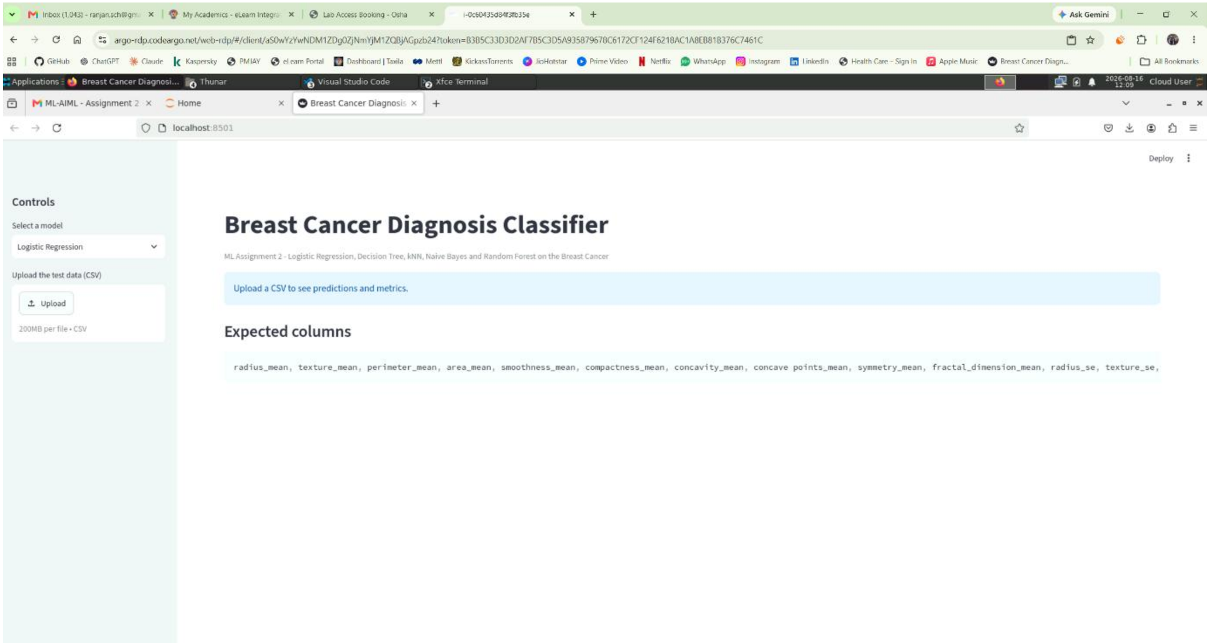

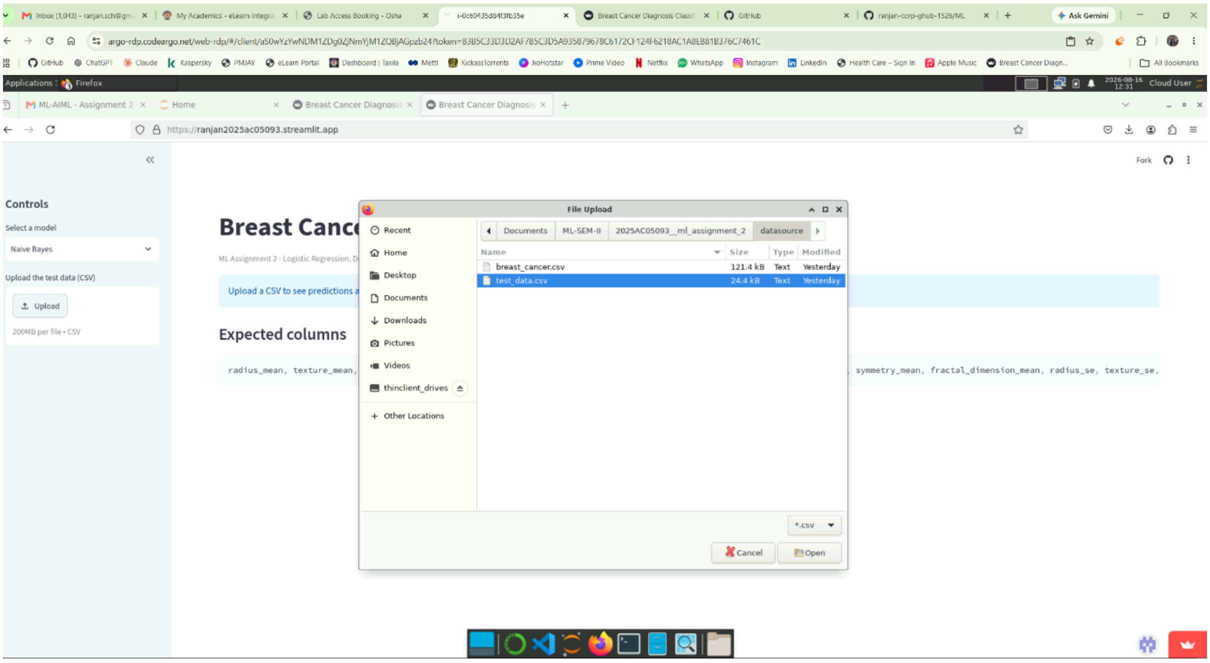

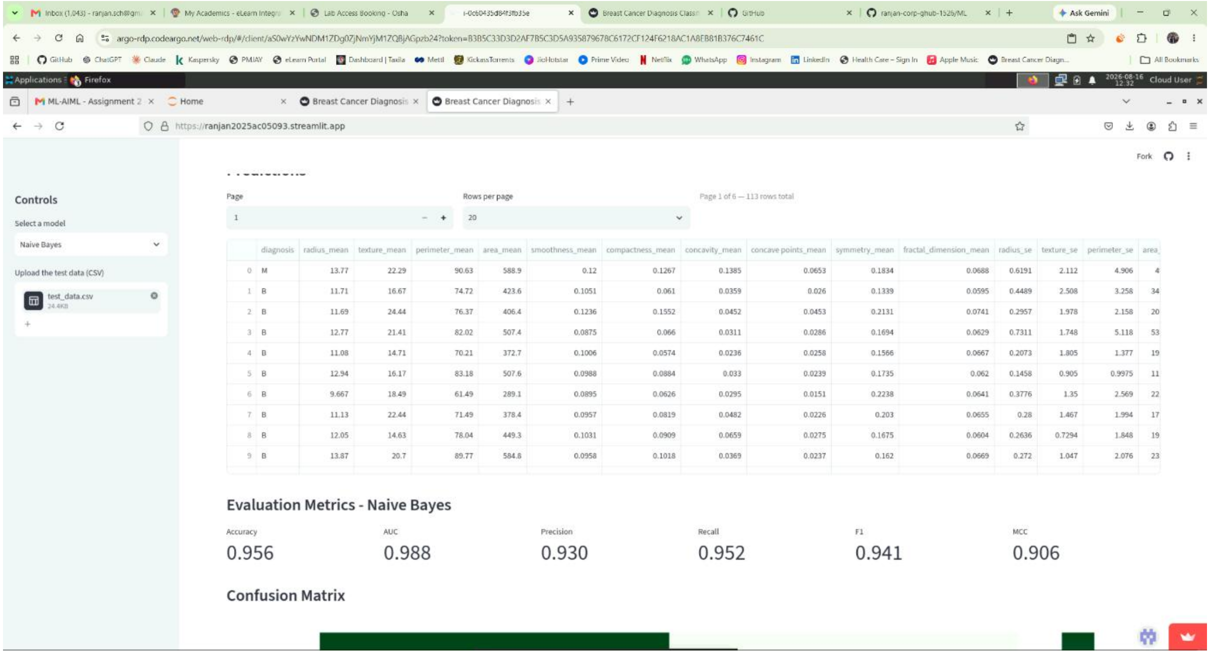

GITHUb ReadMe CONTENT -

Breast Cancer Diagnosis Classification - ML Assignment 2

a. Problem Statement

Early, accurate diagnosis of breast tumours as benign or malignant is critical for treatment decisions. This project builds and compares five supervised classification models that predict a tumour's diagnosis from 30 numeric features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass, and exposes the trained models through an interactive Streamlit app so predictions and evaluation metrics can be explored without touching code.

b. Dataset Description

- *Source:* Breast Cancer Data Set - originally from the UCI Machine Learning Repository, also widely mirrored on Kaggle as [Breast Cancer Wisconsin (Diagnostic) Data Set](#).
- *File used:* `data/breast_cancer.csv`
- *Instances:* 569 (exceeds the 500-instance minimum)
- *Features:* The dataset has 30 numerical features, so it satisfies the minimum requirement of 12 features. These features are based on 10 measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For each measurement, three values are given: mean, standard error (SE), and worst value.

- *Target:* The target column is diagnosis. It contains two classes, M (malignant) and B (benign). So, the problem is a binary classification problem where the model has to predict whether a case is malignant or benign.
- *Class balance:* 357 benign (62.7%) vs. 212 malignant (37.3%) - Did not consider accuracy alone for comparing the models. I also looked at recall for the malignant class and MCC, since predicting a malignant case as benign is more serious.
- *Preprocessing:* all 30 features are continuous and on very different scales (e.g. `area_mean` is in the hundreds, `smoothness_mean` is around 0.1), so they are standardized (`sklearn.preprocessing.StandardScaler`, fit on the training set only) before being fed to every model. The target is label-encoded (`B`→0, `M`→1).
- *Train/test split:* stratified 80/20 split, `random_state=42`. The held out 20% (110+ rows) is exported as `test_data.csv`.

c. GitHub Repository Link

`https://github.com/ranjan-corp-ghub-1526/ML`

- Screenshots uploaded in the Repository

d. Streamlit Link

https://ranjan2025ac05093.streamlit.app/

e. Models Used

Five classifiers are trained on the identical train/test split in `model/train_models.py`:

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbor Classifier
4. Naive Bayes Classifier (Gaussian - fits standardized continuous features naturally)
5. Random Forest Classifier (Ensemble, 200 trees)

Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9649	0.9573	0.9750	0.9286	0.9512	0.9245
Decision Tree	0.9298	0.9246	0.9048	0.9048	0.9048	0.8492
kNN	0.9561	0.9454	0.9744	0.9048	0.9383	0.9058

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Naive Bayes	0.9211	0.9077	0.9231	0.8571	0.8889	0.8292
Random Forest (Ensemble)	0.9649	0.9524	1.0000	0.9048	0.9500	0.9258

Observations

ML Model Name	Observation about model performance
Logistic Regression	Best all-round performer despite being the simplest model — highest recall (92.86%) and F1 (0.951) of all five, tied for highest accuracy (96.49%), and the highest AUC (0.957). Suggests the two classes are close to linearly separable once features are standardized.
Decision Tree	Weakest of the five on almost every metric (accuracy 92.98%, AUC 0.925, MCC 0.849) — a single tree overfits to specific splits and doesn't generalize as well as an ensemble of many trees.
kNN	Strong precision (97.44%, close behind Random Forest) but recall (90.48%) trails Logistic Regression, meaning it misses slightly more malignant cases. Performs well because distance-based methods benefit a lot from the standardized features.
Naive Bayes	Lowest score on every single metric (AUC 0.908, F1 0.889, MCC 0.829, recall 85.71%). Its core assumption — that features are independent given the class — doesn't hold well here, since many tumour measurements (radius, perimeter, area) are naturally correlated with each other.
Random Forest (Ensemble)	Highest precision of all models — a perfect 1.0, meaning zero benign cases were misclassified as malignant on this test set — and the highest MCC (0.926), reflecting the benefit of averaging 200 trees over one. Recall (90.48%) is a touch below Logistic Regression, so it's marginally more likely to miss a malignant case in exchange for never raising a false alarm.
Overall Winner for your dataset?	*Logistic Regression*, by a narrow margin over Random Forest. LR leads on recall, F1, and AUC; Random Forest leads on precision and MCC — genuinely close. In a cancer-diagnosis context, missing a malignant tumor (false negative) is usually costlier than a false alarm, so LR's higher recall tips it as the winner here, though Random Forest is a very reasonable alternative pick if precision/MCC matter more for your framing.